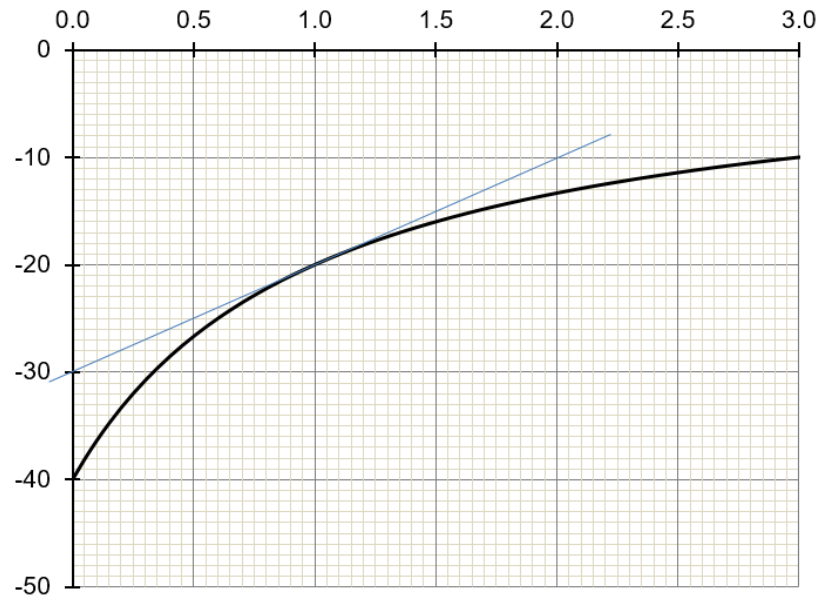


**2 (a)** Gravitational potential (or potential energy) at infinity is zero.

Since gravitational force is attractive in nature, to bring a mass from infinity to a point in the gravitational field, the direction of the external force is opposite to the direction of displacement of the mass. This results in negative work done by the external force

(b) (i)



acceleration = gradient

$$= \frac{[-10 - (-30)] \times 10^6}{(2.00 - 0.00) \times 10^7}$$

$$= 1.00 \text{ m s}^{-2}$$

(ii) Since acceleration is assumed to be constant,

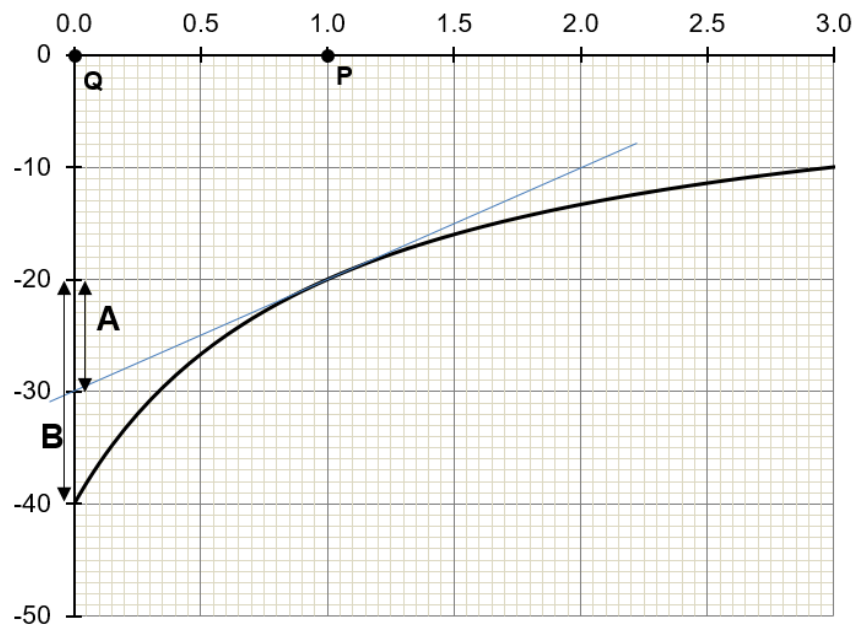
$$v^2 - u^2 = 2as$$

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mas$$

$$= 1.0 \times 1.00 \times 1.0 \times 10^7$$

$$= 1.00 \times 10^7 \text{ J}$$

(iii)



3 (a) The thermodynamic temperature of an ideal gas is proportional to the mean kinetic